[S01] Particle Swarm Optimization (PSO): Optimization Through Collective Swarm Behavior

TOPIC AREA: Swarm Intelligence

I. Learning Objectives and Guiding Question

Explain how a swarm of particles searches for good solutions to continuous optimization problems, examine the effects of algorithm parameters, and compare PSO with local search.

Guiding question: How does PSO combine individual experience and social information to balance exploration and exploitation?

II. Required Tasks

1. Introduce the swarm-inspired motivation, particle population, and the concepts of position x, velocity v, personal best (pbest), and global best (gbest). Distinguish PSO from Hill Climbing.
2. Present the velocity and position update equations; explain ω, c_1, c_2, r_1, and r_2. Manually calculate one update step for two particles in a one-dimensional optimization problem.
3. Select at least two objective functions (e.g., Sphere and two-dimensional Rastrigin); define the search domain and the fitness evaluation procedure.
4. Provide a simple PSO demonstration or a hand simulation with at least five particles over at least five iterations. If coding, compare at least three settings of ω or c_1/c_2 using at least five random seeds.
5. Present a table or plot of the best fitness against iteration count; compare PSO with Random Search or Hill Climbing under the same budget of objective-function evaluations.
6. Discuss premature convergence, parameter sensitivity, and at least two real-world applications (e.g., hyperparameter optimization, scheduling, or engineering design).

III. Equations / Representations to Explain

$$\boldsymbol{v}_i(\boldsymbol{t}+\mathbf{1}) = \boldsymbol{\omega v}_i(\boldsymbol{t}) + \boldsymbol{c}_1\boldsymbol{r}_1(\boldsymbol{pbest}_i - \boldsymbol{x}_i(\boldsymbol{t})) + \boldsymbol{c}_2\boldsymbol{r}_2(\boldsymbol{gbest} - \boldsymbol{x}_i(\boldsymbol{t})); \quad \boldsymbol{x}_i(\boldsymbol{t}+\mathbf{1}) = \boldsymbol{x}_i(\boldsymbol{t}) + \boldsymbol{v}_i(\boldsymbol{t})$$

Student need to define every symbol, specify appropriate units/value ranges and modeling assumptions, and provide a complete step-by-step worked example.

IV. Minimum Expected Outputs

1. Particle states at iterations 0 and 1.
2. A fitness-versus-iteration plot.
3. An analysis of a case in which the method becomes trapped in a local optimum.

V. Required and Recommended Research Papers

Required: Kennedy, J. & Eberhart, R. (1995). Particle Swarm Optimization. Proceedings of ICNN. https://doi.org/10.1109/ICNN.1995.488968

Reading focus: Focus on the swarm concept, pbest/gbest, and examples of function optimization.

Recommended: Poli, R., Kennedy, J. & Blackwell, T. (2007). Particle swarm optimization. Swarm Intelligence 1, 33–57. https://doi.org/10.1007/s11721-007-0002-0

Reading focus: Focus on the roles of algorithm parameters and major PSO variants.

VI. Format and Submission Requirements

- Team size: three students are recommended. Suggested preparation time: approximately two weeks after the topic is assigned.
- Submit a 6–10-page research-style report in Vietnamese on A4 paper (excluding references and appendices), organized as: Abstract, Introduction, Background/Paper Review, Method/Worked Example, Results and Discussion, Conclusion, and References.
- Prepare 8–12 content slides for a 15-minute seminar presentation followed by approximately 5 minutes of Q&A. Every team member must present and answer at least one question.
- Read all papers marked “Required” in this assignment. Explain the material in your own words and distinguish findings reported in the papers from examples or experiments developed by your team.

- Programming-based experiments are encouraged but not generally mandatory. A worked numerical example, an illustrative model, and an evidence-supported comparison are mandatory.
- Cite all figures, tables, datasets, equations, and AI tools used. Independently verify all claims suggested by AI-generated content.

VII. Seminar Assessment Rubric (100 points)

Assessment criterion	Requirements for full marks	Points
Problem statement and objectives	Establish relevant context, a clear research question, and an appropriate scope.	5
Reading and synthesis of research papers	Accurately explain the original papers' objectives, contributions, methods, and findings, with appropriate source comparison.	15
Explanation of algorithms and equations	Define all variables and assumptions, describe algorithmic steps, and provide at least one correct numerical example or diagram.	15
Evidence, comparison, or experimentation	Include an appropriate baseline or reference, verifiable data or examples, and clearly explained figures and tables.	10
Critical analysis and limitations	Discuss strengths, weaknesses, and conditions of applicability. Avoid overstating conclusions based on a small demonstration.	10
Slide quality and citations	Prepare 8–12 clear, legible slides, cite sources for figures/equations and relevant papers, and avoid verbatim copying.	10
Seminar presentation	Deliver a coherent presentation within 15 minutes, explain examples clearly, and ensure balanced participation.	15
Q&A and critical discussion	Answer questions accurately, justify technical decisions, and ensure that each member answers at least one question.	20

Performance levels for each criterion: Excellent (100%), Good (75%), Satisfactory (50%), and Unsatisfactory (0–25%) of the available points, depending on the quality of evidence submitted.